Pasqal Defining the quantum reality

Quantum computing at Pasqal, and its application to the financial sector

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CONFIDENTIAL

Pasqal – A unique blend of science and engineering excellence, with an industrial presence

Pasqal

- Quantum computers based on neutral atoms
- Founded in 2019, at IOGS
- 450+ publications and patents
- 300+ employees, including 75+ PhDs
- Headquartered in France with a global presence – 5+ offices on 3 continents
- 40+ collaborations with industrial partners
- Full-stack approach by integrating Hardware, Software and Applications

- Quantum factories in Paris and Sherbrooke (Canada)
- 2 systems deployed in European HPC centers, 2 more systems to come in 2026

GENCI
Le calcul intensif au service de la connaissance

JÜLICH
Forschungszentrum

Orion Gamma: an industrialized Quantum Computer for datacenter environments

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Standard data center compatible

Setup at room temperature

Rapid scaling: industrial off-the-shelf components

Low energy consumption at 3kW average

How does a neutral atom quantum processor work?

The core components inside a Pasqal QPU

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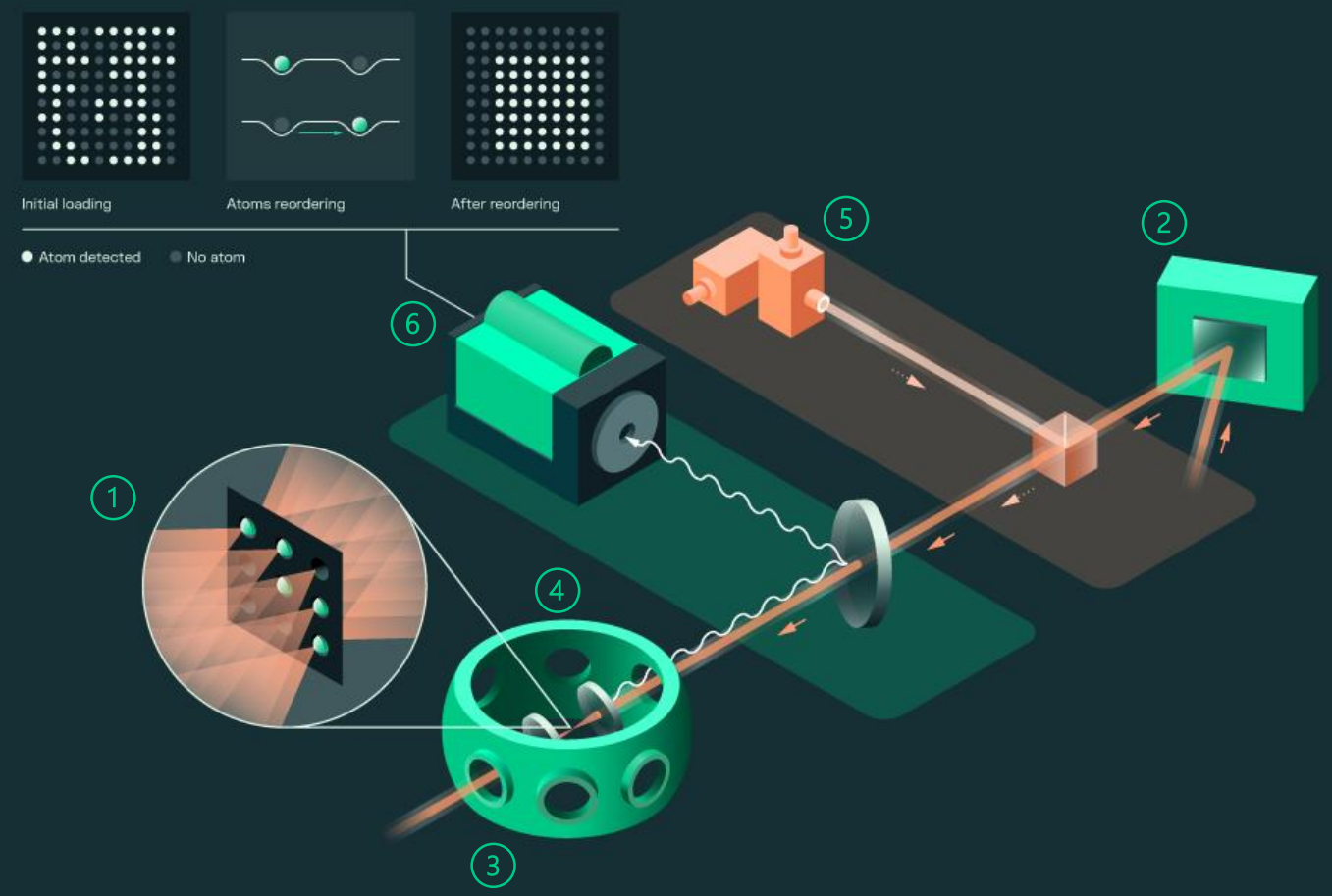

1. Atoms are trapped by a laser
2. Multiple traps are created by splitting the laser with a Spatial Light Modulator (SLM)
3. Vacuum chamber to keep atoms from any disturbances
4. Two high numerical aperture lenses to focus the laser spots
5. Moving atoms is enabled by Acousto-Optical Deflectors (AODs)
6. Readout of the atoms with a fluorescent light detecting camera

Software – how to program a Quantum Computer

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Quantum Applications

Ready-to-use quantum algorithms, designed to address specific computational challenges

Quantum SDK

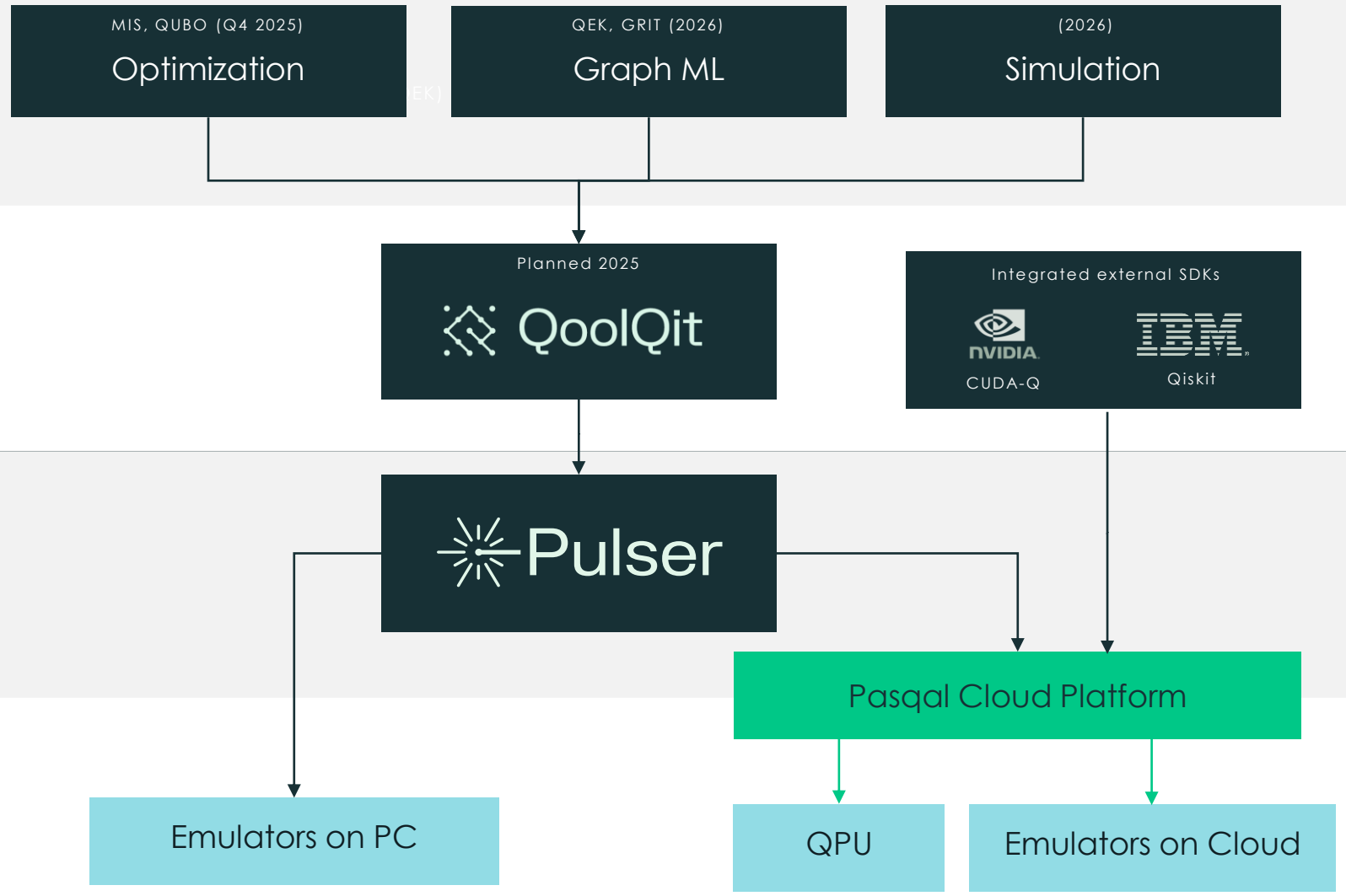
Create custom quantum applications with our intuitive, open-source SDK that simplifies algorithm development or use your favorite external SDK

Neutral Atom Programming

Pulser (and Pulser Studio) provide a flexible ecosystem for programming neutral atom quantum processors, supporting users of all skill levels

Execution

Access Pasqal QPUs and quantum emulators via our cloud platform, and run your jobs



Quantum technology as an integral part of HPC

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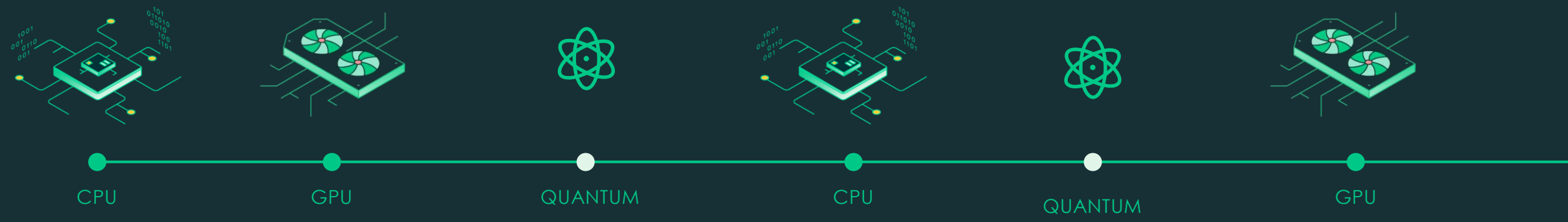
VISUALISATION OF HYBRID COMPUTE WORKFLOW



HPC workflows consist of **hybrid processes** that leverage both quantum and classical computing

Seamless integration with GPU workloads through NVIDIA's CUDA-Quantum

NVIDIA

Standardized QPU job scheduling (SLURM) and support for Qiskit

IBM

Open Access to Orion QPU

Get access to Pasqal's 100 qubits neutral atom quantum computer, "Ruby", hosted at CEA TGCC

WHO CAN APPLY?

EU based researchers in academia and industry

WHAT CAN YOU APPLY FOR?

Up to 100 free hours of quantum computing time

ANY REQUIREMENTS?

One, only open research is allowed

Find out how to access Quantum Computing resources.

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Pasqal Defining the quantum reality

Finance use cases

Predictive Credit Risk Modeling at Crédit Agricole

Pasqal x CRÉDIT AGRICOLE CORPORATE & INVESTMENT BANK

Context

Financial institutions assess debtors' ability to repay loans

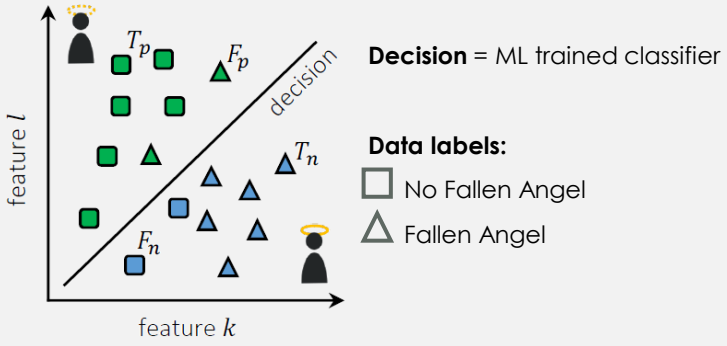
Objective:
Predicting corporate credit rating downgrades (e.g., by agencies like Moody's) offers early insights into potential loan risks

Current method

Crédit Agricole CIB **developed a model to anticipate credit downgrades**, so-called "fallen angels"

Applying a Random Forest Classifier, **a machine learning (ML) technique** to classify a dataset, using so-called decision trees

Trained on a dataset of **65k records** from **2300 companies** in **70 countries**



Quantum solution

A **hybrid classical-quantum** classification algorithm called *QBoost* was proposed

The algorithm was implemented on a Pasqal QPU with **50 qubits**

Result: **>3.5x runtime reduction vs. classical**

Metric	Classical	Quantum	Note
Recall	83%	Same	E.g. out of 100 fallen angels, 83 detected correctly
Precision	28%	Same	E.g. if 100 fallen angels predicted, 28 truly are
#decision trees	1200	50	For quantum equal to #qubits, reduces runtime significantly
Runtime	>3hrs	50mins	

Source: Financial Risk Management on a Neutral Atom Quantum Processor, L. Leclerc et al, arXiv:2212.03223v2

Fraud detection

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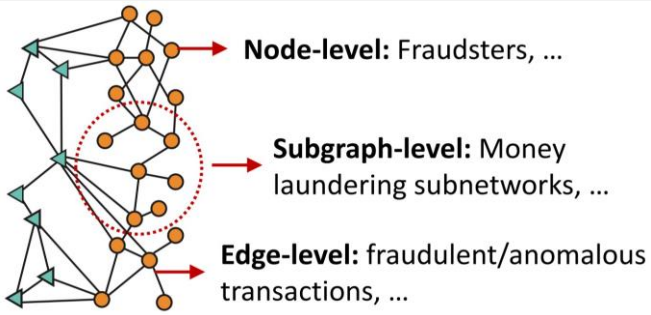
Context

Fraud detection is identifying suspicious activity that indicates criminal theft of money. Commonly performed by analyzing graphs built on transactions within a financial network.

Objective: identify specific features of the graph architecture like suspicious transactions, or nodes representing potential fraudsters.

Problem definition

Malicious actors can collaborate with each other and form networks to run their business successfully. By analyzing network graphs between entities, graph ML can recognize atypical connections or suspicious clusters.

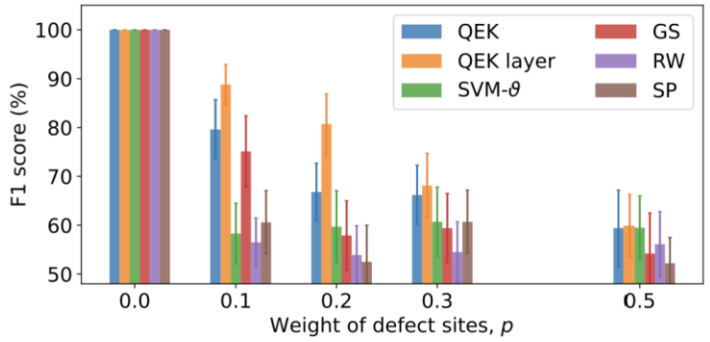


However, classical methods can have troubles grasping global features of such graph organized data.

Quantum solution

Graph machine learning problems can be tackled with Pasqal's proprietary **Quantum Evolution Kernel** (**QEK**) method.

The graph quantum feature map reveals interesting properties: its measured values are influenced by local graph features like node degrees and global ones like cycles, demonstrating the QEK's ability to capture features invisible to the classical kernels.



We also identified other datasets made of bi-partite 2D lattices, for which the quantum procedure exhibited superior performances.

Source: *Quantum Feature Maps for Graph Machine Learning on a Neutral Atom Quantum Processor*

Portfolio optimization

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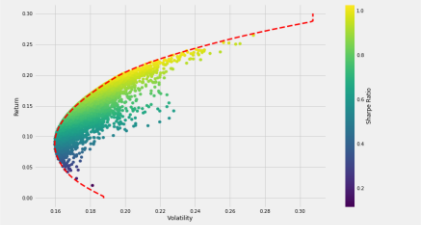
Context

Portfolio optimization is the process of choosing the best mix of financial assets—such as stocks or bonds—to maximize expected return while minimizing risk, based on an investor's goals and constraints.

It involves mathematically balancing potential profits (returns) against uncertainty (risk), often using tools like mean-variance analysis or quadratic optimization.

Problem definition

In its original form, continuous portfolio weights are allowed, i.e. it is assumed that a stock is also tradable 'in parts'. For a more realistic formulation, however, it is necessary to assume the portfolio weight of a stock is 1 or 0, depending on whether the stock is part of the portfolio or not.

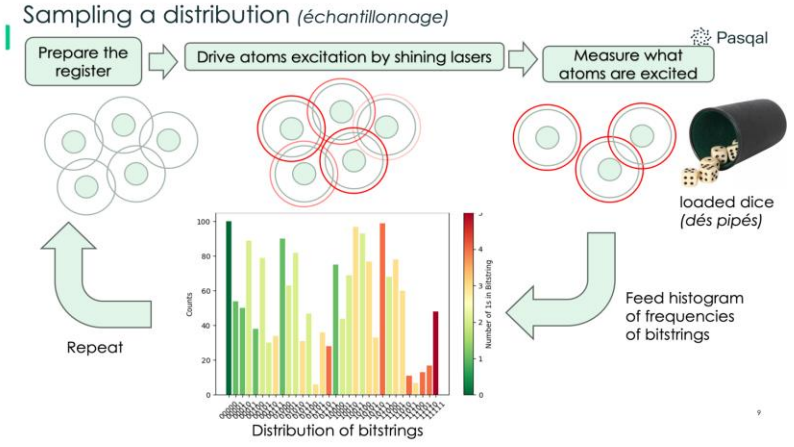


This transforms the problem into something that becomes exponentially complicated in the number of potential stocks, thus hard to solve classically.

Quantum solution

The portfolio optimization problem can be decomposed into a master subproblem that is treated classically and subproblem that is solved thanks to Pasqal's proprietary algorithm of **column generation.**

This hybrid algorithm allows to harness the power of quantum computing by focusing its capabilities on the hard to solve classically subproblem, while the rest of the algorithm is taken care of by a classical machine.



Source:A quantum pricing-based column generation framework for hard combinatorial problems

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Our Roadmap

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HARDWARE R&D

Physical qubits	200	1,000		10,000		
Base repetition rate	1 Hz	3 Hz		10 Hz		100+ Hz
Control features	Z-add	Zoning	In-circuit lossless readout	In-circuit reloading	Fast decoders	Large-scale local gates
			10+ PIC enabled parallel 1Q gates		100+ PIC enabled parallel 2Q gates	
					Full addressability	
Logical qubits		2		20		200
Logical fidelity		95%	98%	99,9%		99,9999%
				Break-even ✕		MegaQuOps ✕

HARDWARE PRODUCTS ON-PREM & CLOUD

Orion			Vela	Centaurus	Lyra
Alpha 100+ qubits	Beta 100+ qubits	Gamma 140+ qubits	250+ qubits First QA	Analog & Early FTQC	Analog & Impactful FTQC

ALGORITHM & ACCELERATED LIBRARIES

Hybrid Quantum/Classical computing		Slurm integration	Hybrid programming model development			Low-latency HW co-location
Quantum application		QA on Quantum Matter		QA on Optimization & ML		
Open-source libraries		GML apps	Optimization applications	Specialized libraries for Q Simulation, Optimization and Graph ML		Digital FTQC SDK and applications

2022 2023 2024 2025 2026 2027 2028 2029 ›

How does a neutral atom quantum processor work?

Encoding and processing quantum bits, or qubits

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Atom loading

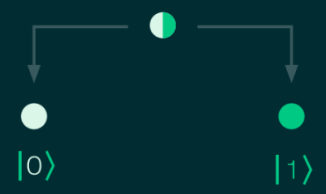
A laser acts like an optical tweezer, trapping atoms

Multiple traps are generated by splitting the laser beam

Half of the traps will contain an atom due to stochastics

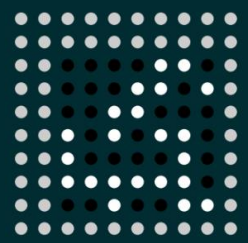
Register preparation

Atoms can be moved to create custom grids, forming the quantum register



Quantum processing

Manipulating the array of atoms with other lasers allows for quantum calculations by switching atoms between energy states



Results read-out

Final results are read by imaging the array: excited atoms (ones) are dark, ground-state atoms (zeros) are bright